

Lalit Maurya

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PROFILE

Computer Science undergraduate with experience in full-stack development and AI/ML, skilled in Rust, Python, and modern web technologies. Passionate about building scalable, real-world solutions that empower individuals and organizations.

EDUCATION

Shiv Nadar Institution of Eminence

Bachelor of Technology in Computer Science and Engineering (CGPA 8.39)

Delhi NCR, India

Aug. 2023 – Expected May 2027

EXPERIENCE

Technology Consultant Intern

Microsoft

May 2025 – July 2025

Hyderabad, India

- Designed and shipped a 5-agent AI architecture (Azure OpenAI, .NET 8/C#, Power Automate, Dataverse) that automated seller pre-meeting prep and post-meeting CRM updates by translating direct seller feedback into agent workflows, cutting RFP turnaround from 3–5 days to under 1 minute and saving sellers 5+ hrs/day; published to Microsoft's internal AI Hub for org-wide reuse.
- Built a sales enablement platform surfacing real-time stakeholder insights, next-best-action recommendations, and auto-generated executive decks from live enterprise data, engineered with a layered service architecture, Azure DevOps CI/CD, human-in-the-loop review, and cost-optimized inference, shaped through continuous iteration with sellers and stakeholders to fit their actual workflow rather than a generic pipeline.

Hackathon Director, @ping

Google Developer Group

April 2024 – Feb 2025

Delhi NCR, India

- Created and led @ping, a campus-wide hackathon initiative, organizing hackathons across diverse formats and bootcamps that helped students progress from beginner to intermediate development skills.
- Led the initiative end-to-end, driving event ideation, technical direction, reusable infrastructure and libraries, bootcamp curriculum, and execution, including a Connect Four bot framework that enabled participants to build, test, and compete with their own AI agents.

Teaching Assistant - Computer Organisation and Architecture

Shiv Nadar Institution of Eminence

Aug 2025 – Nov 2025

Delhi NCR, India

PROJECTS

oxigrad-autograd | Rust, Python, PyO3, Maturin

Jul 2025

- Developed a scalar-valued auto-differentiation engine in Rust, implementing forward/backward passes and gradient computation through dynamic computation graphs; Published to PyPI with Python 3.9+ support, enabling seamless installation via `pip` or `uv`
- Built Python bindings with PyO3 and Maturin, integrating common activation and loss functions including Sigmoid, ReLU, Softmax, MSE, and Cross Entropy

hive | Go, Python, Zig, Kubernetes, Docker

Feb 2026

- Designed a distributed framework coordinating LangGraph agents across Kubernetes, enabling parallel execution and dependency-aware synchronization while optimizing for developer ergonomics
- Built near-clickless deployment and integration workflows, including containerization and SSH-based node bootstrapping that auto-joins servers to the cluster and transparently pauses graph nodes until required data is available

Skin Detection Without Color Information | OpenCV, Scikit-learn, NumPy

Feb 2025 - May 2025

- Implemented grayscale skin detection using facial priors, texture features, and region growing on a HELEN-derived dataset with ground-truth segmentation masks
- Extended the WACV 2017 approach by jointly incorporating multiple complementary features for skin classification, rather than evaluating each feature independently

QueenSweep | Rust, Web Assembly, TypeScript, Chromium Extension

Oct 2025 - Dec 2025

- Built a constraint satisfaction solver for LinkedIn Queens with aggressive pruning via 1-step lookahead and pluggable heuristics, solving 90% of puzzles in <10ms (p90: 9.55ms, 250+ states/ms)
- Engineered a zero-copy Rust solver with shared immutable state and WebAssembly bindings, then integrated it into a Chromium extension using DOM mutation observers and synthetic events for automated solution application

TECHNICAL SKILLS

Languages: Rust, JavaScript, TypeScript, Python, C, C++, SQL, HTML, CSS, C#

Frameworks & Libraries: React, Next.js, SvelteKit, Flask, FastAPI, PyTorch, Poem-OpenAPI, .NET 8/10

Tools & Platforms: Git, GitHub, Supabase, Docker, Google Cloud Platform (GCP), Azure, Azure Foundry, Linux, uv

Databases & ORMS: MySQL, PostgreSQL, MongoDB, Prisma, SQLx, Dataverse

Technologies & Protocols: Node.js, REST APIs, gRPC, WebSockets, Selenium